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ABOUT ME

I am a proactive Software Engineer and a student at the National University of Science and Technology POLITEHNICA Bucharest, within the Faculty of Industrial Engineering and Robotics. My academic background in robotics complements my technical expertise in building complex digital systems, such as my flagship project Garison—a 2D top-down RPG featuring a custom systems-architecture with GPU-accelerated shaders and high-speed serialization.

Currently, I serve as a Programming Mentor at Logiscool. Teaching Python and algorithmic logic has honed my ability to distill technical concepts into clear, accessible knowledge.

WORK EXPERIENCE

 **Logiscool** – Bucharest, Romania

Programming mentor

[01/10/2025 – Current]

- * Mentorship & Communication: Simplify complex programming logic (Python, Algorithmic thinking) for students of varying skill levels.
- * Classroom Leadership: Managed diverse groups of [2 to 20] students, maintaining high engagement levels while balancing different learning speeds and technical backgrounds.
- * Logic Optimization & Troubleshooting: Conducted real-time code reviews and live-debugging sessions, identifying and resolving logical errors in student-led projects.
- * Curriculum Adaptation: Translate abstract computer science fundamentals into practical, visual projects.

PROJECTS

[01/06/2022 – Current]

Garison - Solo Developer A technically ambitious 2D top-down RPG developed over four years of iterative refactoring and systems engineering. The project serves as a comprehensive case study in scaling a monolithic script-base into a modular, high-performance engine.

* Iterative Architecture Evolution: Executed a 4-year roadmap of continuous system overhauls, transitioning from simple event-driven scripting to a sophisticated, decoupled framework built from scratch.* Modular Systems Architecture: Engineered a robust Entity Finite State Machine (FSM) to manage complex AI decision-making and interaction logic, ensuring 100% decoupling from physics and collision modules.* High-Performance Persistence: Architected a JSON & Binary buffer serialization engine capable of deep world persistence, allowing for instantaneous saving/loading of complex memory states and DS Map structures.* GPU-Accelerated Shader Engineering: Developed a custom GLSL fragment shader pipeline for atmospheric rendering, managing real-time day/night cycles and dynamic lighting effects with zero CPU overhead.* Lexical Narrative Engine: Built a custom string parser and dialogue controller featuring real-time tag extraction for dynamic text effects (shake, wave, color) and interactive branching narrative logic.* Object-Oriented Inventory Logic: Designed a constructor-based inventory system utilizing modern GMS 2.3+ structs for item stacking, dynamic slot allocation, and scalable item-attribute management.* State-Driven World Progression: Implemented a global Quest Controller and state-map system that manages complex cross-room progression and persistent environmental triggers.

Link: <https://github.com/hexa325/Garison2023>

SKILLS

Skills

Advanced AI-Augmented Workflows (BMad Methodology)

* Structured AI Engineering: Deep practical knowledge of integrating Large Language Models into the software

lifecycle via the BMad framework.

- * Multi-Agent Orchestration: Capable of managing complex agent interactions (Party Mode, Advanced Elicitation) to force critical review, adversarial problem-solving, and robust technical planning.
- * Lifecycle Control: Strict adherence to phase-gate development—ensuring Business alignment (WDS-1) and UX validation (WDS-4) before initializing Agentic Implementation (WDS-5).
- * Context Engineering: Expert in managing technical memory and context injection (e.g., GEMINI.md, project-context.md) to guide AI agents in producing accurate, project-specific code without hallucination.

Full-Stack Web Development

- * Core Architecture: Next.js (App Router, Server Components, SSR/SSG), React, TypeScript, JavaScript (ES6+).
- * Backend & Data Integration: Node.js (via Next.js Server Actions & API Routes), RESTful API Architecture, JSON Data Modeling, Server-Side Data Fetching.
- * Styling & UI Engineering: Tailwind CSS v4, shadcn/ui, Advanced CSS3 (Animations, Custom Utilities), Semantic HTML5.
- * State & Performance: React Server/Client State Management (Hooks, Context), Build Optimization, Web Vitals & SEO Configuration.

Systems & Game Programming

- * Languages: Python (Advanced/Mentoring level), GML (GameMaker Language), C/C++ (Academic/Robotics).
- * Graphics: GLSL (Fragment/Vertex Shaders), GPU-accelerated rendering.
- * Architecture: Finite State Machines (FSM), Object-Oriented Programming (OOP), Data Serialization (JSON/Binary Buffers).

Engineering & Robotics

- * Hardware/Logic: Industrial Automation, Control Systems, Robotics Systems Architecture.
- * Tools: MATLAB, CAD (Basic), Logic-based programming.

DevOps & Tools

- * Version Control: Git, GitHub (Branching, PRs, Actions).
- * Deployment: Vercel, GitHub Pages.